



Multi-modal Zebra Crossing Detection Framework for Visually Impaired Navigation

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Abstract

Independent walking poses a significant challenge for visually impaired individuals, especially during street crossings. The absence of visual cues like pedestrian lights and directional signs leads to mutual blind spots between pedestrians and nearby vehicles, increasing the risk of collisions. This necessitates advanced assistive technologies to ensure pedestrian safety and improve mobility. This research proposes a Multi-modal Zebra Crossing Detection Framework for Visually Impaired Navigation (MZCDFVIN). At its core is the YOLOv11 Hough Line Transform Convolutional feature-enhanced Ship Rescue Cross-Attention (YHTCSCA) model, designed for real-time obstacle detection and zebra crossing recognition. To address training issues such as non-convergence, instability, and vanishing gradients, the Ship Rescue Optimization Algorithm (SROA) is introduced. For improved image processing and signal detection, the framework incorporates the Haze U-Net with LeCun Network Modular Fully Convolutional Network (HUNet-MC), combining Haze U-Net for image enhancement and LeNet for traffic light recognition. The system achieved high performance across evaluation metrics: 99.79% accuracy, 99.82% precision, 99.81% recall, 99.76% F1-score, 99.3% mean Average Precision (mAP), and 99.5% Intersection over Union (IoU). These results highlight the framework's reliability and efficiency in supporting safe street-crossing for individuals with visual impairments. The combination of the YHTCSCA model with the HUNet-MC framework offers a strong solution for navigation for visually impaired pedestrians. By precisely detecting the zebra crossings and recognizing traffic signals, the system enhances street-crossing safety, reduces blind spot risks, and supports greater pedestrian independence. This multi-modal approach establishes a new benchmark for assistive navigation technologies in smart urban environments.

Keywords Visually impaired and blind · Zebra crossing detection · Traffic sign recognition · Obstacle detection · Safe pedestrian mobility

Introduction

The visual impairments provide assistive devices like white canes serve as foundational elements for technologies like the guide cane and the miniguide. A white cane aids persons

with blind has identifying nearby obstacles and navigating their environments [1]. One of the most significant factors impacting outdoor navigation for blind individuals is the presence of stressors. Although advancements in technology aimed at addressing these stressors have been made, many of these systems necessitate substantial infrastructural modifications, like the installation of tactile paving or the use of expensive equipment [2, 3]. Nevertheless, the challenge faced by the visually impaired community is their difficulty in navigating the external environment safely and independently, as outlined in the report on blind navigation and the role of technology.

While various hardware-based solutions have been suggested, including wearable devices and systems that depend on external servers for data processing, these options often entail considerable financial implications. The expenses

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associated with such products, coupled with the necessity for continuous server operation, render these solutions impractical, especially for Visually Impaired Blind (VIB) individuals in developing nations where resources are scarce [4–7].

Recent advances in object and zebra crossing detection used to aid individuals with visual impairments through Machine Learning (ML) and Deep Learning (DL) have enhanced the experience of road crossing for blind persons. A specific TinyML model has been developed to detect obstacles on pavements aimed at supporting blind people who have difficulty navigating through urban environments [8]. This approach leveraged an Enhanced Inverted Residuals (EIRs) block and a Lightweight Vision Transformer (ViT) block [9]. This approach develops a novel model to demonstrate an affordable and easy solution for VIB people to cross roads safely, to detect the presence of Pedestrian Traffic Lights (PTL) and zebra crossings, and to provide information about the traffic light and onward crossing direction.

Many researchers have identified changing traffic signals, pedestrian crossings, and obstacles in a dynamic environment for safe navigation. Khan et al. [10] developed a DL transfer-based Object Detection Model (ObDtM) for a customized dataset that included the obstacles that the VIB recognized were obstacles that considered threats. Nonetheless, several challenges arise, particularly concerning the absence of dependable knowledge and movement aids, which were often prohibitively expensive. The approach has been developed specifically by Ashiq et al. [11] for VIPs to facilitate their flexibility and enhance their protection. The application employs the MobileNet architecture, chosen for its minimal computational demands, allowing it to operate effectively on low-power devices. However, the primary obstacles faced by VIB include object detection and recognition, identification of currency, interpretation of textual information, translation, as well as mobility, navigation, and safety concerns. The integration of Artificial Intelligence (AI) and Edge Computing (EC) technology was suggested to improve the security of pedestrians with VIB. An RGB-D image sensor was innovated by Chen et al. [12] to capture data regarding the surrounding environment, which was then processed by an EC module and communicated to visually impaired pedestrians via auditory alerts. However, it was essential to tackle the challenges posed by complex traffic conditions and safety concerns that visually impaired pedestrians face while navigating. Joshi et al. [13] introduced an innovative wearable assistive device powered by AI, known as AI-SenseVision. Additionally, the integration of various sensors, cameras and advanced technologies has led to significant improvements; it has also increased the overall cost of the device. An advanced guiding system that operates via a smartphone. Chen et al. [14] employed a MobileNet

model that has been fine-tuned through transfer learning for feature extraction on overlapping grids. For the detection of Tactile Walking Surface Indicators (TWSI), which utilize the Single Shot multibox Detector (SSD). However, it was challenging to maintain the magnitude and caliber of the dataset, which easily resulted in overfitting issues.

Jeong et al. [15] designed a system that utilizes the You Only Look Once version 8 (YOLOv8) DL model to identify walkable zones, public transportation amenities and obstacles, providing appropriate guidance to the user. Nevertheless, there were difficulties in merging computer vision with navigation functionalities. Li et al. [16] introduced an unsighted difficulty recognition classification utilizing the Partial Convolutional-YOLO (PC-YOLO) model and incorporating a Partial Convolutional Feed-forward Network (PCFN), which aims to minimize computational redundancy. Nevertheless, intricate street environments continue to present considerable challenges for visually impaired difficulty recognition systems. Atitallah et al. [17] proposed the creation of a different hindrance recognition system utilizing a sophisticated YOLO v5 neural network. Nevertheless, the obstacle avoidance tasks present significant challenges due to a variety of factors, like changing traffic sign conditions, substantial obstruction and both intra-class and inter-class variations. Wang et al. [18] designed a novel obstacle detection approach for visual navigation support, referred to as YOLO-OD. This developed the Feature Weighting Block (FWB), which enhances the insight into characteristic significance. Consequently, the significant challenge encountered by individuals with visual impairments was mobility, as they often find it difficult to recognize and navigate around obstacles, especially in urban areas. Atitallah et al. [19] demonstrated the detection system utilizing an adapted YOLO v5 neural network architecture. Despite advancements, VIB individuals continue to encounter numerous challenges in their daily activities. A wearable assistance system utilizing vision technology to enhance situational awareness for persons who were unsighted or blind in indoor environments. Li et al. [20] constructed the non-intrusive wearable devices, which include a Red Green Blue-Depth (RGB-D) camera, an Embedded Computer, and Haptic Feedback Modules (ECHFM). Nevertheless, the immediate environment leads to diminished mobility for individuals with visual impairments. Bouteraa [21] suggested navigation system was fundamentally based on sensors, processing units, a Decision Support System utilizing Fuzzy Logic (DSSFL), and a user interface. Nevertheless, in outdoor environments, users of the system encountered some collisions owing to the unpredictable nature of the obstacles. Chen et al. [22] designed a wearable movable device for individuals with visual impairments by participating in semantic visual Simultaneous Localization And Mapping (SLAM) with a recently

developed advanced platform. Santos et al. [23] utilized an RGB-D camera along with Haptic Feedback (RGBDHF), all incorporated into a specially designed backpack aimed at enhancing usability and user interaction. However, the focus was primarily on functionality, overlooking other critical factors that affect usability, such as comfort and ease of use. A methodological framework was proposed to extract participants' gazing behavior and estimate visual attention from eye-tracking data, including eXtreme Gradient Boosting (XGBoost) and SHapley Additive exPlanations (SHAP) were designed by Jiang et al. [24] used to model and understand how geometric features like distance, incidence angle, pitch and sign size contribute to specific gaze behaviors and allowed distinguishing between focusing and scanning behavior. However, despite the benefits of high information density, there was a drawback in terms of the stability of participants' moving speed. An advanced model called Enhanced Convolutional Elongated Herd Lightweight Yolo-fastest V2 (ECEHLY-V2) was developed by Jemin et al. [25]. The model was implemented and designed to provide increased functionality and ease of user navigation through environments with obstacles. Current issues include identifying obstacles, which affect orientation due to low availability of landmarks, user navigation was reliant on others, and there was difficulty understanding the visual data provided in the new environment. The semantic segmentation Convolutional Neural Network (CNN) proposed by Son et al. [26] was built for low-compute devices and also has competitive performance with 71.2 mIoU on Cityscapes and 74.46 mIoU on Camvid. This runs at 209 Frames Per Second (FPS) with 480932 image resolution on a Giga Texel shader eXtreme (GTX) 1080 desktop and 50 FPS on the low-compute Jetson. But in many situations, real-time performance was indicated by maximum FPS on the desktop and not either this mobile computing platform. Kan et al. [27] designed a eLabrador, a proof-of-concept wearable navigational system intended to help the visually impaired walk long distances while exploring natural and built outdoor environments. However, the outdoor spaces were complex because of the variety of obstacles, traffic signals and rules, and social customs that create significant challenges and obstacles for the mobility of visually impaired people, especially for long distances. Table 1 presents an examination of the existing methodology.

Problem Statement

Numerous obstacles continue to exist in the realm of zebra crossing detection for individuals with blind. The obstacles include the significant costs linked to assistive technologies, the difficulties in integrating computer vision with navigation systems, and the challenges of dataset scalability that

lead to overfitting. Additionally, elements such as complex traffic situations, varying lighting conditions, occlusion, and the complexities of urban settings hinder the precision of obstacle detection. While advanced DL models like YOLO and MobileNet improve detection performance, ensuring dependable mobility and navigation in urban environments continues to be a major challenge. Fig. 1 illustrates the challenges associated with traffic light and zebra crossing detection.

To mitigate these dangers, an intelligent object detection system is needed to detect CMOs, alert VIPs and enhance navigation assistance. Thus, the system significantly reduces the likelihood of accidents, promoting safer, independent movement for VIB.

Motivation

The motivation for developing a thorough object detection and traffic awareness system for individuals with visual impairments stems from the urgent need to enhance safety and autonomy during daily navigation. Individuals with vision impairments experience unique challenges when recognizing moving objects and determining the danger associated with motor vehicles, bicycles, and pedestrians, especially in busy or poorly managed intersections. Conventional traffic signal systems often do not meet the needs of persons who are blind or have low vision, and depend on infrastructure that is not equally or always available or well-adapted for individuals who have vision challenges. This system enables VIB individuals to move to outdoor surroundings with superior safety and confidence.

Contributions

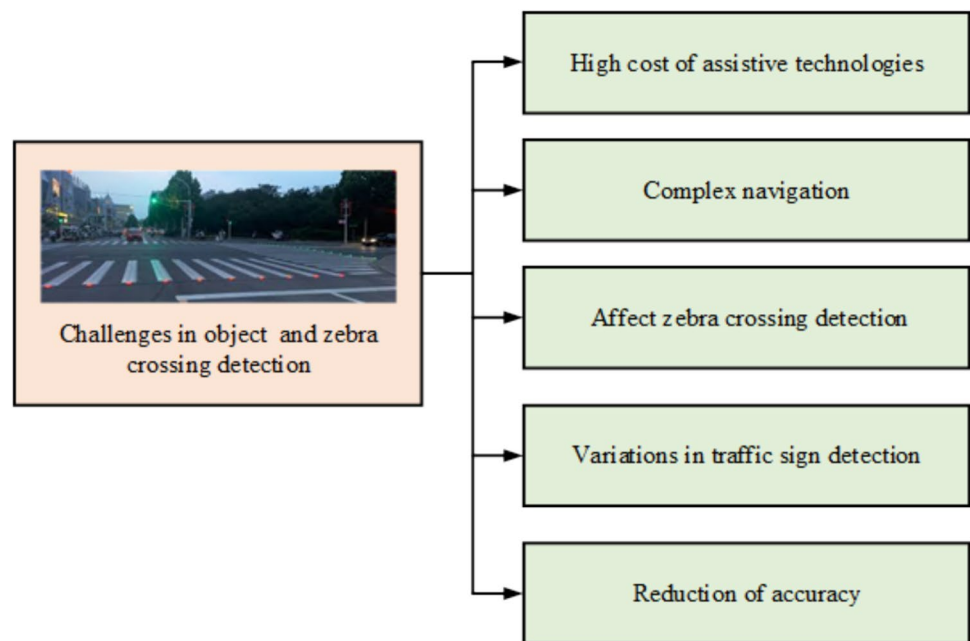
- A new technique known as MZCDFVIN is used for the traffic crossing detection system and YHTCSCA helps blind users isolate challenges and categorize zebra crossings to ensure safe navigation. YHTCSCA uses YOLOv11 to recognize vehicles, moving pedestrians and possible obstacles, assuring users possess full situational awareness.
- The Hough Line Transform (HLT) detects the linear characteristics of zebra crossings by recognizing white stripes and analyzing their alignment and slope.
- Alternatively, a CNN model trained on labeled images accurately identifies zebra crossing patterns across various conditions, thereby enhancing safety and precision in navigation.
- The suggested network undergoes fine-tuning through the application of the SROA. To address issues related to non-convergence failure, training instability and

Table 1 Evaluation of the existing approaches

References	Major components	Types	Object detection	Pros	Cons	Performance measures
[10]	ObDtM	Wearable devices	DL-based object detection	ObDtM provides obstacle identification and Improved mobility	ObDtM has higher energy consumption	• Accuracy:92.5% • Precision:90.8% • Recall: 91.2% • F1-score: 91.0%
[11]	MobileNet architecture	Handled devices	CNN-based object recognition	MobileNet architecture demonstrates high detection accuracy and operates effectively in various lighting conditions	This encounters difficulties when faced with occlusions	• Accuracy:83.3% • F1-score: 91.1%
[12]	RGB-D image sensor	Wearable devices	AI-based guide system	RGB-D image sensor provides intelligent scene perception and adaptive learning	Challenges in consistent updates	• Accuracy:85.8% • F1-score: 93 ms
[13]	AI-SenseVision	Wearable devices	ML-based object detection	AI-Sense Vision provides economical and immediate processing	Limited computational power	• Mean Average Precision (mAP): 81.34% • F1-score: 82.07%
[14]	MobileNet with TWSI and SSD	Smartphone-based	AI-based object detection	MobileNet was compatible with smartphones	Challenges in reliance on the processing capabilities of mobile devices	• Accuracy: 93.76% • Sensitivity: 93.61% • Specificity: 94.03%
[15]	YOLOv8	Smart glasses	YOLOv8 based on object detection	YOLOv8 provides rapid processing capabilities and accommodates practical applications	Challenges in high computational expenses	• Precision: 86.6% • Recall: 82.4% • F1-score: 84.3%
[16]	PC-YOLO	AI-based navigation system	PC-YOLO object detection	PC-YOLO improves exceptional accuracy and was tailored for applications	PC-YOLO requires high-end hardware	• Precision: 83.7% • Recall:79.7% • mAP: 84.1%
[17]	YOLO v5 neural network	Embedded AI and Smart devices	DL-based obstacle detection	YOLO v5 enhances the energy-efficient and compact design of the network	Limited processing speed	• Clock frequency: 2500 MHz • Speed:54 m/s
[18]	YOLO-OD with FWB	AI-based guide system	YOLO-OD object detection	YOLO-OD improves high-speed obstacle recognition	Challenges in hardware-intensive resources	• mAP: 30.02% • Accuracy:90.4%
[19]	YOLO v5	Wearable devices	DL-based obstacle detection	YOLO v5 enhances scene comprehension and capabilities	Limitations in dynamic environments	• mAP: 76.41% • FPS: 9.4 s
[20]	ECHFM	Vision-based cognitive assistance	Depth and color image processing for 3D environmental mapping	Non-intrusive design; provides situational awareness in indoor scenarios	Limited to indoor environments; potential challenges in dynamic settings	• Detected accuracy: 90.78% • FPS: 10 s
[21]	DSSFL	Sensor-fusion-based navigation aid	Ultrasonic sensing combined with fuzzy logic for obstacle detection and path guidance	Real-time feedback combines haptic and auditory cues and adaptable decision-making	Causes body fatigue due to constant hand interaction; limited to predefined scenarios	• Precision: 78.34% • Recall: 89.67%
[22]	SLAM	Vision-based navigation with semantic SLAM	Real-time semantic visual SLAM integrating CNN for object recognition and mapping	High-precision object recognition, effective in complex environments and operates on low-cost devices	Requires substantial computational resources; potential limitations in dynamic or crowded environments	• Accuracy: 89.97% • Predicted time: 78.56%

Table 1 (continued)

References	Major components	Types	Object detection	Pros	Cons	Performance measures
[23]	RGBDHF	Obstacle detection wearable device	Sensor-based detection	Provide an accurate obstacle detection wearable device	Challenges in the specific imbalanced dataset information	• Accuracy: 79.45% • Precision: 82.78%
[24]	XGBoost and SHAP	VR-based Wayfinding	Not primary focus (focus on gaze & attention tracking)	Reveals pedestrian visual behavior, improves urban design	High cost, requires VR setup, limited to simulations	• Visual attention accuracy: 89.6% • Gaze tracking precision: 90.78%
[25]	ECEHLY-V2	Real-time Scene Perception	YOLO-based detection for obstacles	Real-time obstacle detection, adaptable to various environments	Sensitive to lighting, requires high processing	• Accuracy: 91.67% • Latency: 90.56 ms • F1-score: 89.78%
[26]	CNN	Infrastructure-enabled Navigation	Basic object detection (landmark recognition)	High reliability in structured environments, low latency	Needs infrastructure support, less effective outdoors	• Navigation success rate: 89.67% • Latency: 78.89 ms • User satisfaction: 90.67%
[27]	eLabrador	Wearable Navigation System (eLabrador)	YOLO/Faster R-CNN for object & obstacle detection	Wearable, real-time guidance, multi-modal feedback	High energy consumption, costly	• Detection accuracy: 91.45% • Precision-Recall: 90.56% • FPS: 9.7 s • Power efficiency: 89.56%

Fig. 1 Challenges of the zebra crossing detection

vanishing gradient phenomena for the detection of the zebra crossing.

- A novel approach utilizing HUNet-MC employs Haze U-Net for the removal of haze and leverages the LeNet architecture for the recognition of traffic signs.
- The pre-processing stage enhances image quality and prepares the data for precise detection. The Recursive Mean Separate Histogram Equalization (RMSHE) technique improves contrast under varying lighting situations.

This manuscript is divided into multiple sections. Section "Proposed methodology for object detection and zebra crossing" outlines the methodology utilized for object and zebra crossing detection, whereas Section "Experimental results analysis" describes the experimental methods applied and assesses the resulting data. Section "Final remarks" provides a comprehensive summary of the findings. Lastly, Section "Conclusion" delivers a concise overview of the conclusions reached and proposes potential avenues for future research.

Proposed Methodology for Object Detection and Zebra Crossing

This section enhances the detection of traffic signals and navigation of zebra crossings present considerable difficulties for individuals with visual impairments, particularly in bustling urban settings. Conventional approaches often fall short in providing precision and dependability, especially

under conditions of low illumination or when views are obstructed. Fig. 2 illustrates the overall proposed methodology for object detection and zebra crossing.

This addresses these issues by utilizing advanced DL methodologies, including YOLO11 for object detection, Hough transforms for recognizing crosswalks and CNN for precise identification of traffic signals, thereby facilitating safer navigation.

Data Collection

The data gathered from the PTL (<https://github.com/samuel-yu2002/ImVisible?tab=readme-ov-file>) is a comprehensive image dataset focused on street intersections, specifically designed for the identification of PTLs and zebra crossings. The dataset includes images that exhibit a range of variations in weather conditions, positioning, and orientation concerning the traffic lights and zebra crossings, as well as differences in the size and type of intersections.

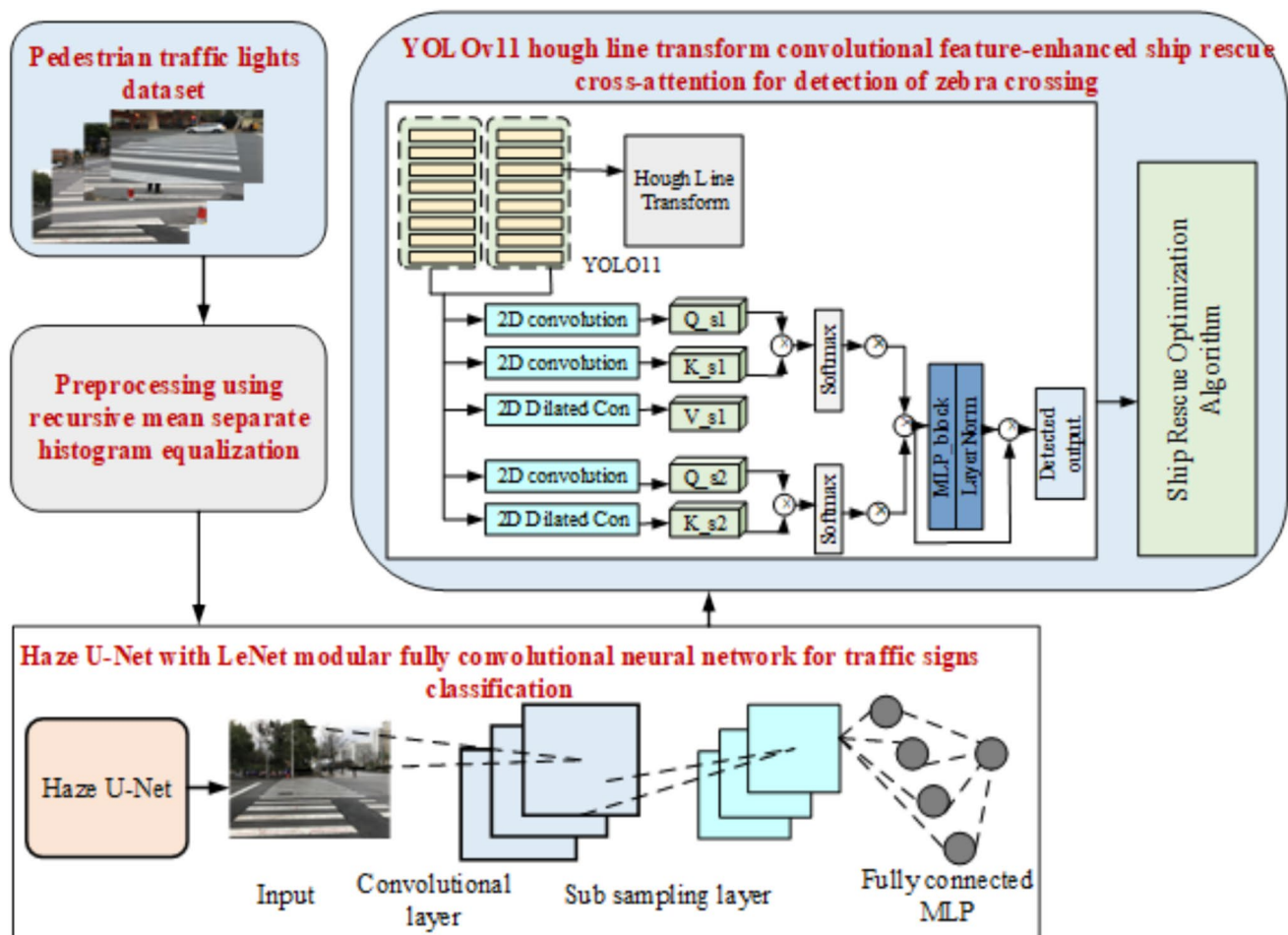


Fig. 2 Workflow of proposed methodology

Preprocessing

After the completion of data collection, preprocessing methods are employed on the collected images to minimize variations in brightness and enhance the contrast of the object and zebra crossing detection. RMSHE is the histogram that is initially divided in a manner akin to Basic Histogram Equalization (BHE). Subsequently, each segment is further divided into two distinct parts in a recursive manner. This process is applied to enhance images of objects and traffic light images using RMSHE is evaluated as in Eq. (1).

$$F(X) = Y_n + [Y_G - Y_n]/2^n \quad (1)$$

where, $F(X)$ represents the transformation applied to pixel intensity X , Y_n is the histogram of the input image divided by its average intensity value and Y_G denotes that the recursive function implements a distinct transformation. Overall, RMSHE enhances the quality of images and readies the data for precise detection and contrast under diverse lighting conditions. After the preprocessing techniques, the image quality expanded, subsequently leading to the traffic sign detection process utilizing the HUNet-MC approach.

Haze U-Net with LeNet Modular Fully CNN for Classifying Traffic Signs

This section describes the HUNet-MC approach to effectively enhance contrast by uniformly redistributing distorted values according to the histogram for the recognition of traffic signs. This approach involves the Haze U-Net in conjunction with a LeNet-based Modular Fully CNN (LeNet-MFCN). The resulting contrast-enhanced feature map consists of three channels. This eliminates haze from the input image [28, 29]. The elimination of haze is frequently grounded in the atmospheric scattering model and is calculated as in Eq. (2).

$$I(y) = K(y) t(y) + B(1 - t(y)) \quad (2)$$

where, $I(y)$ represents the hazy image, $K(y)$ denotes the haze-free image, B signifies the global environment light and $t(y)$ refers to the communication map, which illustrates the quantity of light that arrives at the camera. Figure 3 represents the process of HUNet-MC for classifying the traffic lights.

This collective data serves as the input for the second-stage neural network, ultimately producing a clear image devoid of haze known as the Haze Reduction Module

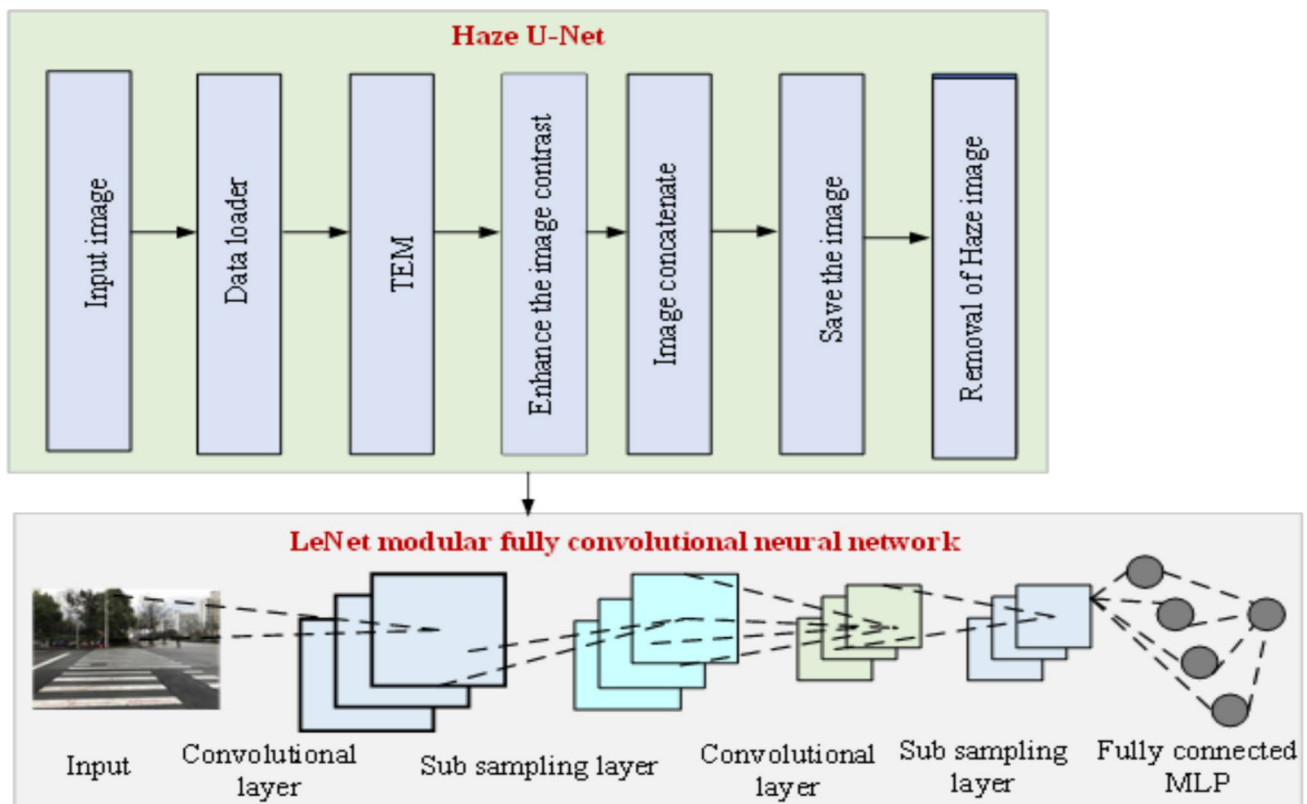


Fig. 3 Traffic light classification using HUNet-MC

(HRM). Depending on the specific implementation, the Transmission map Estimation Module (TEM) and image contrast enhancement stages are executed concurrently. The outcomes are combined with the contribution of the image and transmitted to the HRM, which efficiently eliminates the haze $K(y)$ employed as in Eq. (3).

$$K(y) = \frac{I(y) - B}{t(y)} + B \quad (3)$$

where, the Haze U-Net model is designed to estimate the transmission map $t(y)$ and the atmospheric light, B enabling it to reconstruct a clear image $K(y)$. The synergy of these two components guarantees precise recognition and localization of traffic signs, even under difficult conditions like low light, haze, or fog. The Histogram of Oriented Gradients (HOG) algorithm is employed to characterize the distribution of image gradients across various orientations, effectively capturing shape and aspect features for the classification of traffic signs as evaluated in Eqs. (4, 5)

$$M(a, b) = \sqrt{H_a^2 + H_b^2} \quad (4)$$

$$\theta(a, b) = \tan^{-1}\left(\frac{H_b}{H_a}\right) \quad (5)$$

where, $M(a, b)$ signifies the intensity of the gradient, while $\theta(a, b)$ indicates the direction of the gradient. LeNet-MFCN generated from vertical strip-pooling is expressed as in Eq. (6).

$$H_m^{ts} = \max_{0 < j < g} H_{nyg+jm} \quad (6)$$

where the strip-pooled feature maps directions are denoted as H_m^{ts} , respectively. Each of these pooled features is processed through a 1×1 convolutional layer that contains 512 filters, which is subsequently followed by a ReLU activation function. The quantity of learnable parameters remains constant, while the receptive field of the filters is expanded by interspersing intermediate kernel values with zeros as evaluated in Eq. (7).

$$F_j = \sum_k F[j + s \cdot l] * w[l] \quad (7)$$

where, s specifies the dilation rate of the kernel with weight w , which is applied to the input tensor F through convolution. Standard convolution is considered a specific instance of atrous convolution, where the rate s is set to 1. The gradient orientations are categorized into discrete bins, like nine bins corresponding to the range of 0° to 180° . The voting

process for each bin is influenced by the magnitude of the gradient, denoted as $M(a, b)$ is evaluated in Eq. (8).

$$B_l = \sum_{j \in C} N_j \cdot w_l(\theta_j) \quad (8)$$

where, the weighting function, denoted as $w_l(\theta_j)$, specifies the extent to which each pixel contributes to the corresponding orientation bin. Subsequently, a Support Vector Machine (SVM) classifier is implemented to eliminate false positives and is evaluated as in Eq. (9).

$$f(y) = w^T y + a \quad (9)$$

where, the weight vector is denoted as w^T , the HOG feature vector is represented by y and the bias term is indicated by a . In the classification phase, LeNet-MFCN is designed to improve the technique's evaluation, enabling the classification of traffic signs for the identification of VIB into specific categories are calculated using Eq. (10).

$$Y'_{i,j} = \sum_x \sum_y Y_{j+ni+m} \cdot L_{m,n} \quad (10)$$

where, Y represents the input image, L denotes the convolutional kernel and Y' signifies the resulting characteristic map. Finally, predictions are made using the traffic signs identified during the classification phase to evaluate the effectiveness of the prediction of traffic flow signs for VIB in detecting and classifying traffic signs.

Translation Using Text-To-Speech Converter

After identifying the traffic signs, the information is transformed into audio signals using a Text-To-Speech (TTS) system. This method vocalizes the names of identified objects to aid visually impaired individuals in effective navigation. The system retrieves the relevant object labels (e.g., "Stop Sign" and "Green Light") from the classification outputs from both LeNet-MFCN and YHTCSCA, and transfers the labels to a TTS engine, such as Google TTS, for audio output. Before the labels are converted, the output contains some text preprocessing, like formatting, context-sensitive messages and removing any unnecessary alerts. After the TTS performs converting the textual labels into intelligible speech, the intelligible speech is relayed to the user through headphones, announcing detected traffic signals and revealing their position on the detected route and instructing navigation. An example is to inform the user with instructions like "For green". The traffic signal is green. You are now allowed to move through the intersection safely." The

multimodal feedback provides blind users with opportunities to transfer complex graphic information into functional and practical audio instructions. Speech output is delivered to users in headphones. Thus, Haze U-Net is employed to effectively eliminate haze, thereby improving visibility, while the LeNet-MFCN is responsible for sensing and categorizing traffic signs. Once the classification of the traffic signs is completed, the detection of the zebra crossing is performed using the YHTCSCA approach.

YOLOv11 Hough Line Transform Convolutional Feature-Enhanced Ship Rescue Cross-Attention for Detection of Zebra Crossing

This section introduces the YHTCSCA technique, calculated to assist blind individuals by sensing obstacles and identifying zebra crossings to ensure safe navigation. By employing YOLOv11, the system is capable of recognizing vehicles, pedestrians, and potential dangers, thus offering comprehensive situational awareness.

The functionality of the HLT within the YHTSCA framework acts as a complementary component as it assesses geometrical properties of zebra crossings such as alignment, slope, and spacing of the white stripes; therefore, this two-part system is useful for cases where YOLOv11 detects a zebra crossing incorrectly or fails to detect it at all based on obstructions or fading of the zebra crossing. The HLT improves identification consistency by confirming linear patterns, which help address issues associated with variable conditions. This is proven by producing measurable improvements to detection accuracy or robustness, particularly in settings with low-light or if the visual conditions are cluttered, depending on YOLOv11 or optimizing the HLT integration to sustain computational efficiency. Fig. 4 illustrates the procedure of the YHTCSCA technique.

Furthermore, the CNN Enhanced Cross Attention Mechanism (CNN-ECAM) model, which is trained on labeled images, effectively recognizes zebra crossing patterns across different conditions, thus enhancing safety and accuracy in navigation. The pixels of the image are allocated to the

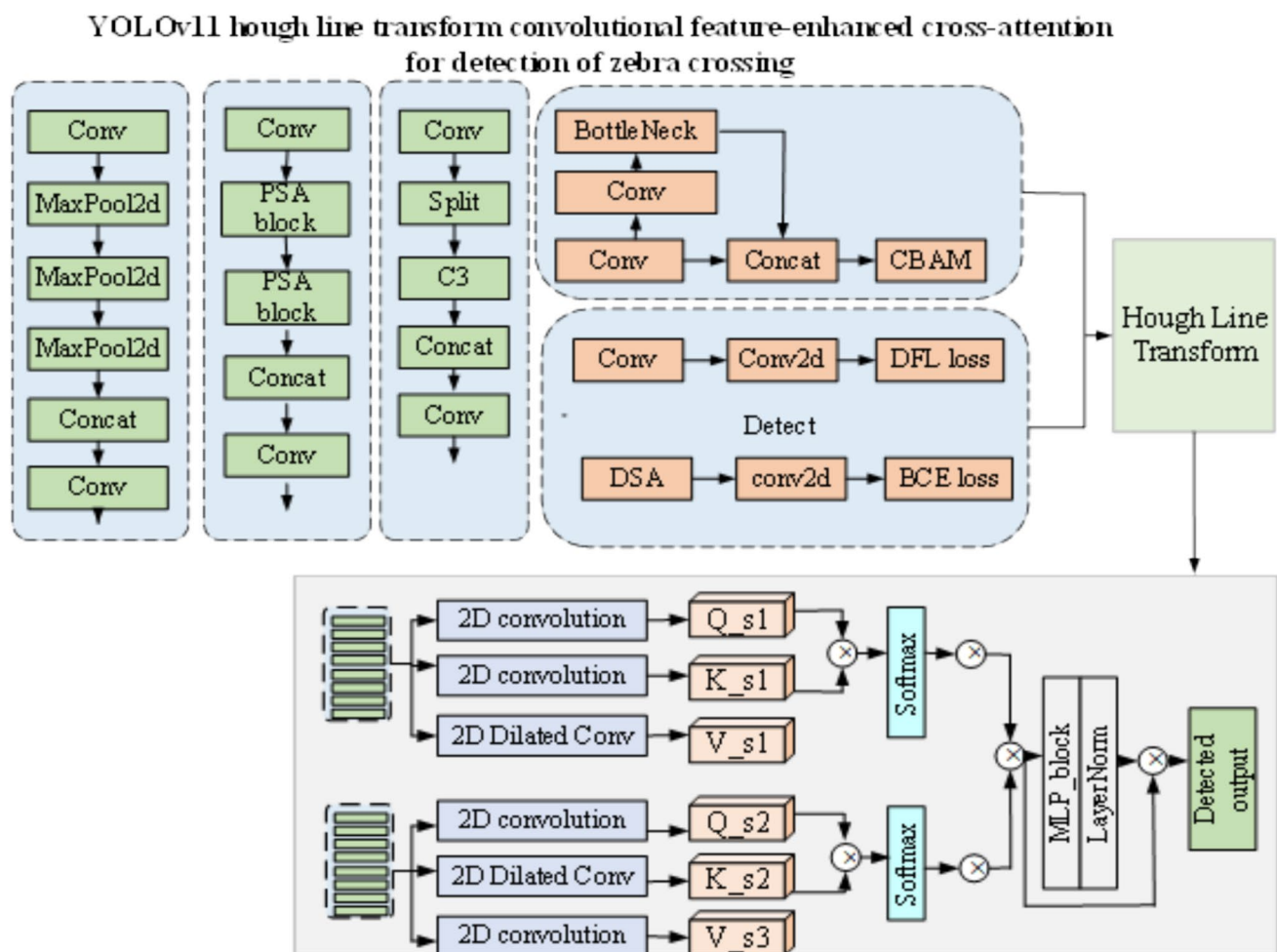


Fig. 4 Structure of the YHTCSCA method

parameter space. First, train spatial features from labeled images, extracting pixels using the convolutional layers of the CNN. Then, apply the cross-attention mechanism. For example, cross-attention strengthens attention to areas like zebra patterns by evaluating the relationships among pixel regions. The pixel data is then mapped to parameter space, where important features, in this case, lines that signify zebra stripes, are identified by some accumulation scores, such as the Hough Transform. This process allows a confident and accurate detection of zebra crossings, regardless of lighting and environmental conditions, thus enhancing navigational safety. In the modified parameter space, lines are subsequently identified using accumulation scores [30–32]. The Parallel Spatial Attention (PSA) block enhances the differentiation of spatial features by employing attention mechanisms across multiple spatial scales. The Dilated Spatial Attention (DSA) block effectively models contextual dependencies with dilated convolutions, enabling us to grow the receptive field without growing the number of constraints. The Convolutional Block Attention Module (CBAM) is a simple mechanism that improves the representation of features via channel attention followed by spatial attention. HLT employs the subsequent expression to collect the targets along the specified line is calculated as in Eq. (11).

$$|s_l - y_i \cos \theta_l - x_i \sin \theta| \leq \gamma S \quad (11)$$

where, γS represents the maximum deviation, which is determined based on the deviations of the targets within the group. The determination of the linearly distributed group targets is expressed in Eqs. (12, 13).

$$\overline{G}_0 : \delta_l = \{(a_i, b_i, c_i) \in G_0\}^{\widehat{M}}_{i=1} \quad (12)$$

$$\overline{G}_1 : \delta_l = \{(a_i, b_i, c_i) \in G_1 \text{ or } G_0\}^{\widehat{M}}_{i=1} \quad (13)$$

where, $\overline{G}_0$ and $\overline{G}_1$ denotes the scenarios in which the linearly distributed group targets are used to detect patterns, respectively. To identify each pattern, the detection probability is calculated as in Eq. (14).

$$P^{S1}_R(t) = 1 - \sum_{\widehat{M}_k + \widehat{M}_k \leq 2} q(\widehat{M}^1_k | \overline{G}_1) q(\widehat{M}^2_k | \overline{G}_0) \quad (14)$$

where, M signifies the total quantity of potential lines within the image, which corresponds to the dimensionality of the parameter space associated with the HLT. Under the condition of $\widehat{M}$, the distribution becomes more intricate due to the presence of true targets being detected at the zebra linear characteristics of zebra crossings by detecting white stripes and evaluating their alignment and

slope. YOLOv11 construction involves three principal elements: the backbone, the neck and the head. The backbone serves as the main feature extractor, employing CNNs to convert the original image into a series of feature maps across various scales, thereby capturing essential structural details. This module is structured with multiple heads. For $Y \in H^{R \times T \times C}$, the associated query Q , key K and value V are projected accordingly. The Sliding Window Dual Attention (SWDA) module combines dual attention mechanisms through both standard and dilated convolutions within sliding windows, facilitating improved local–global contextual awareness. This module significantly boosts detection accuracy, particularly in scenarios involving occlusion or clutter. The channels of the feature map are segmented into n distinct parts and executed across various heads utilizing different null rates calculated as in Eqs. (15, 16).

$$g_i = SWDA(Q_i, K_i, V_i, s_i), 1 \leq i \leq m \quad (15)$$

$$Y = Linear(Concat[g_1, \dots, g_n]) \quad (16)$$

where, g_i denotes the output of the SWDA module for the i – th head, while s_i signifies the dilation rate associated with the i – th head. This emphasizes that $N_c(f)$ is calculated through the element-wise summation of the descriptors obtained from both Global Average Pooling (GAP) and Global Max Pooling (GMP), thereby facilitating the information through a multilayer process is evaluated as in Eq. (17).

$$N_c(f) = \beta [MLP(GAP(f)) \oplus MLP(GMP(f))] \quad (17)$$

where, $N_c(f)$ denotes the input characteristic map, $GAP(f)$ is the process that performs global pooling by calculating the average value across all spatial locations within each feature channel. $GMP(f)$ is method is akin to GAP, $\oplus$ is the output from the MLP applied to $GAP(f)$ and $GMP(f)$ are concatenated, enabling the model to leverage both average and max-pooled representations and β is a non-linear function. The final output characteristic map is obtained through the element-wise summation of the input feature maps provided as in Eq. (18).

$$G_0 = G_i \oplus G_{HR} \quad (18)$$

where, G_0 is the feature representation resulting from the processing stage. G_i is the preliminary feature representation before any transformations are applied and G_{HR} denotes the enhanced feature representations achieved through a specific transformation, like feature enhancement, attention mechanisms, or residual connections. The transformer demonstrates strong abilities in extracting feature

information, allowing it to identify long-range dependencies and gather global contextual insights. Following the acquisition of these tensors, it performs element-wise multiplication to generate improved features B_1 and B_2 that have been processed through the attention mechanism and are expressed as in Eqs. (19, 20).

$$B_1 = \text{softmax}\left(\frac{Q_1(KR(S_1))}{\sqrt{d_{K_1}}}\right)V_2 \quad (19)$$

$$B_2 = \text{softmax}\left(\frac{Q_2(KR(S_2))}{\sqrt{d_{K_2}}}\right)V_1 \quad (20)$$

where, d_{K_1} represents the dimension of K_1 , while d_{K_2} denotes the dimension of K_2 . The resulting summed features are then processed through a multi-layer perceptron block that employs a residual structure expressed using Eq. (21).

$$D = L_c[MLP[B_1 \oplus B_2]] \oplus (B_1 \oplus B_2) \quad (21)$$

where, MLP refers to the multi-layer perceptron, while L_c stands for loss function. The architecture of the MLP primarily consists of two rectilinear layers, disconnected by the Gaussian Error Linear Unit (GELU) stimulation function. This approach helps visually impaired people navigate by detecting obstacles and identifying zebra crossings. Although YOLOv11 excels at identifying zebra crossings, the information varies under the circumstances. There is a possibility that YOLO misclassifies or fails to detect crosswalks in certain scenarios, including faded road markings, low-light conditions, shadows, occlusions, or environments with significant visual clutter. The HLT is not merely a redundant tool; instead, its capability to identify geometric line shapes enables the system to validate YOLO's predictions and ensure that the shapes of the identified crosswalks remain consistent (e.g., parallel stripes) and correctly oriented. This integration minimizes the false positives and false negatives is crucial for applications related to the safety of pedestrian and/or visually impaired navigation for real-time performance. To minimize the loss function parameter, SROA is used as fine-tuning, giving an improved solution to detect obstacles and identify zebra crossings.

Ship Rescue Optimization Algorithm

This section provides the proposed SROA, which is included in the approaches made to improve the YHTCSCA method's loss parameters. The ship rescue process is divided into two cases: a first case with a known specific spot where the person in distress is located and the second case is a search in a confined area, giving the fastest resolution possible [33].

• Initialization

This initialization of the number of objects and their locations is presented as Eq. (22).

$$Y_i(t+1) = Y_{\min} + s \times (Y_{\max} - Y_{\min}) \quad (22)$$

where, t represents the iteration time, s signify a casual number within the range of 0 to 1. The position of the ship i at the t -th iteration is indicated by $Y_i(t)$. Additionally, $Y_{\min}$ and $Y_{\max}$ denotes the boundaries of the problem for the detection of the objects and the zebra crossing.

• Fitness Function

The Fitness Function represents the optimization procedure described in Eq. (23). This approach aims to diminish the loss function to improve the detection of objects and zebra crossings.

$$F(x) = \{\min\{loss\}\} \quad (23)$$

where, $F(x)$ denotes the fitness function and $\min\{loss\}$ represents the objective of reducing the loss parameter associated with the detection of objects and zebra crossings. Figure 5 presents the main steps involved in the proposed SROA methodology.

Thus, YHTCSCA significantly improves navigation safety for individuals with visual impairments. This methodology accomplishes its goals by identifying obstacles, recognizing zebra crossings, and assessing potential hazards. Furthermore, it offers reliable situational awareness across diverse environments, significantly improving accessibility and mobility.

The pipeline of RMSHE for preprocessing, U-Net for dehazing, LeNet-MFCN for classification, YOLOv11 for object detection, HLT for geometric analysis, and a CNN-based cross-attention with optimization involves considerable computational complexity and latency. Including the SROA expedites the application while still being cost-effective. The composer SROA optimally modifies crucial parameters throughout the detection and optimization process and efficiently directs search decision-making. This is analogous to establishing priority lines when rescuing people from a disaster. When paired with hardware acceleration and streamlined model variants, SROA optimizes the multi-stage pipeline RMSHE, U-Net, HLT and YHTCSCA, enhancing its responsiveness and practicality for implementation. As a result, the integration of SROA preserves high detection accuracy and reliability and boosts system efficiency, rendering the solution more economical and applicable for real-world scenarios.

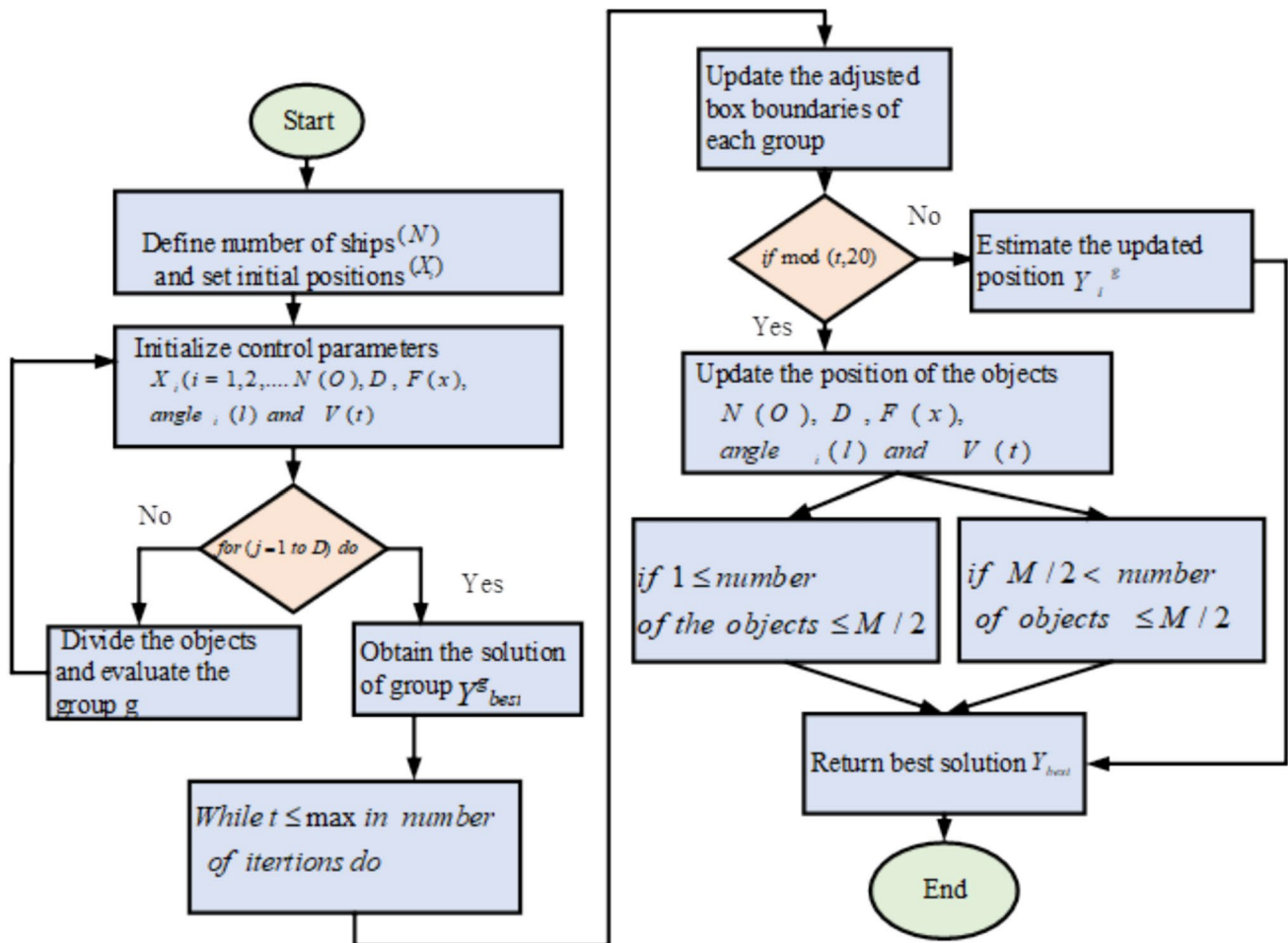


Fig. 5 Flowchart of the SROA

Experimental Results Analysis

This section outlines the quantitative outcomes obtained from the YHTCSA methodology, which has been created to facilitate the identification of traffic lights, objects, and zebra crossings utilizing the Python programming language. Additionally, the assessment of zebra crossing identification also tracks convergence analysis and the fitness function. This approach demonstrates its potential usefulness for visually impaired individuals in future assistive applications.

Table 2 presents a comprehensive review of the parameter settings applied during implementation, highlighting critical configurations for training the YOLOv11 model in Python. The model is set to handle input images 640×640 pixels, either trained with the Adam optimizer or SGD with the learning rate of 0.001. The eight adjustments for the RGB frames to just one learning rate help with each learning rate to reduce samples by requiring input images on the low end of the learning scale. The dropout rate between the

training session varies from 0.2 to 0.5 per session to ensure not lose samples per training epoch to help prevent overfitting. The batch size is maintained at 32 to facilitate effective gradient updates, and the Leaky ReLU activation function is utilized to introduce non-linearity and prevent vanishing gradients. The training phase extends over 100 epochs to improve overall performance.

Table 2 Analysis of parameter settings

Parameters	Values
Implemented tool	Python
Base architecture	YOLO11
Input size	640×640
Optimizer	Stochastic Gradient Descent (SGD)/Adam
Learning rate	0.001
Drop rate	0.2–0.5
Batch size	32
Activation function	Leaky ReLU
Epochs	100

Dataset Description

This section describes the Pedestrian-Traffic-Lights (PTL) dataset, focused on street intersections, specifically designed for the identification of pedestrian traffic signals and zebra crossings.

Table 3 outlines a dataset comprising 3,849 images, allocated with 80% designated for training and 20% for testing, each associated with a single label. An 80:20 split is a widely accepted benchmark; it ensures the model has enough data, 3,079 images, to identify patterns while having enough unseen data to accurately evaluate its generalization capabilities. The image resolution is 768×576 pixels, with mean values [120.57, 119.17, 113.85] and standard deviation [66.32, 65.09, 65.68] for normalization.

Analysis of Detected Outcomes

This section defines the ground-truth labels for the detection of the zebra crossing and traffic light. This examines the detected images and assesses each image by applying the method for the detection of zebra stripes and traffic lights of VIB individuals.

Figure 6(a, b) illustrates a comparison between the original images and their respective classifications of traffic signals. The left column features original images depicting pedestrian crossings, whereas the right column presents the classification outcomes, indicating the detection of traffic signals labeled as "green" or "red." This classification system aids visually impaired individuals in safely determining when it is appropriate to cross. The system proficiently identifies traffic lights and accurately assigns labels, showcasing its capability for providing navigation.

Figure 7(a-c) illustrates the methodology for detecting zebra crossings through a YHTCSCA technique. The initial column features the original images taken at street intersections. The subsequent column presents images with detected patterns, where edge detection methods, including the HLT, emphasize the zebra crossing patterns in red. The concluding column showcases the final detected images, in which the system effectively recognizes and annotates the zebra crossing area with bounding boxes and textual labels.

Table 3 Evaluation of dataset descriptions

Dataset parameters	Values
Total number of images	3849
Training images	80%
Testing images	20%
Number of labels per image	1
Image size	768 × 576
Mean	[120.57, 119.17, 113.85]
Standard deviation	[66.32, 65.09, 65.68]

This comprehensive approach guarantees precise detection, thereby improving pedestrian safety by facilitating navigation support for individuals with visual impairments.

Figure 8 shows the detection of zebra crossings under various contexts. In subfigures, Fig. 8(a-b) show the original and detected zebra crossing during the daytime, where the model highlighted the crossing area. In subfigures, Fig. 8(c-d) show the nighttime detected crossings, and each instance is detected successfully, even in low illumination. The last set of subfigures, Fig. 8(e-f) shows the original and slippery road results, where the framework respectively detected the zebra crossings and showed the detection and surface hazards. Overall, the figure aims to show the robustness of the detection framework in terms of lighting and road conditions.

Performance Analysis

The assessment of the presentation system involves the classification and detection of the traffic signs and zebra crossing, which compares current methodologies like ObDtM [10], MobileNet architecture [11], RGB-D sensor [12], AI-SenseVision [13] and the newly proposed YHTCSCA.

Figure 9(a-f) present data comparing a range of techniques utilizing performance metrics including accuracy, precision, recall, F1-score, mAP and IoU. Among the methods assessed, the proposed YHTCSCA demonstrates superior performance, achieving an accuracy of 99.72%, precision of 99.82%, recall of 99.81%, F1-score of 99.76%, mAP of 99.3%, and IoU of 99.5%. This performance significantly surpasses that of other models such as ObDtM, MobileNet, RGB-D with AI, and AI-Sense. The perfect scores of YHTCSCA underscore its effectiveness in detecting zebra crossings, establishing it as the most dependable method for pedestrian safety applications.

Figure 10 illustrates the data comparing a range of techniques utilizing performance metrics, including execution time and training or testing time. The execution time denotes the duration required to process an individual image or frame during inference at runtime. Conversely, the testing time generally signifies the cumulative duration taken to assess the model across the entire test dataset or the average per sample, encompassing preprocessing, inference and postprocessing. Fig. 10(a) illustrates that the proposed YHTCSCA consistently achieves a low and stable execution time of approximately 7 to 8 ms, whereas obDtM [10] experiences a significant increase from 6 ms to 29 ms, as the input size AI-Sense [13] remains steady at around 11 to 12 ms, RGB-D with AI [12] increases from 6 ms to 18 ms, and MobileNet [11] rises from 6 ms to 17 ms. Fig. 10(b) depicts the duration of training and testing over 100 epochs, showing that both metrics increase from approximately 21

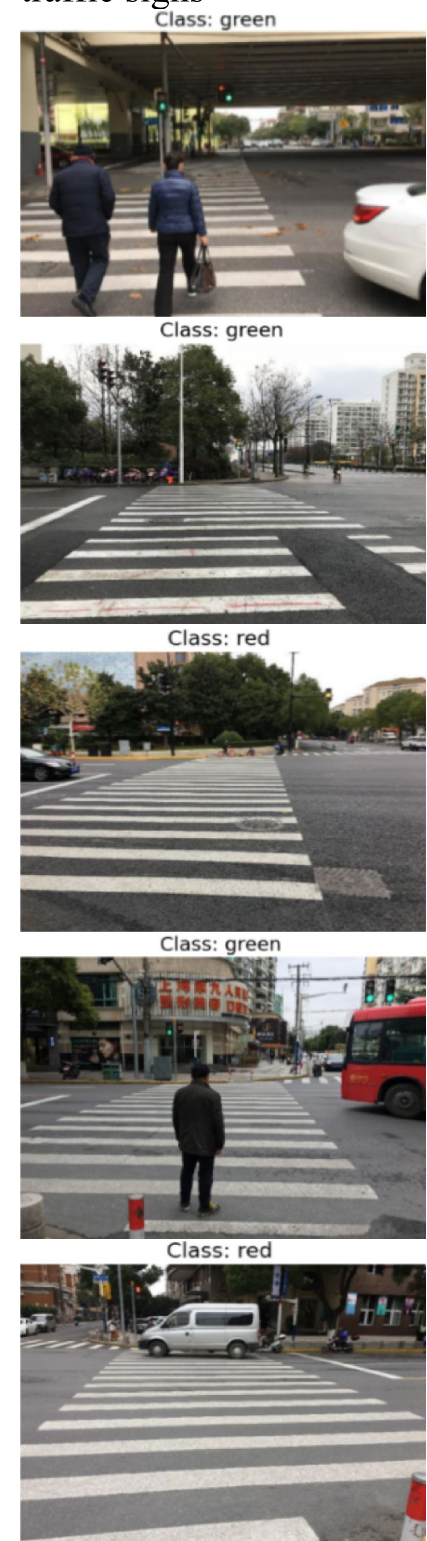
Fig. 6 Outcomes based on (a) Original images and (b) Traffic sign detection

Original image



(a)

Classification of the traffic signs



(b)

Fig. 7 Evaluation based on (a) Original images, (b) Pattern detected images and (c) Final detected images



seconds to about 32 seconds by the 25th epoch. As the training continues, the times begin to level off, with the training duration reaching a maximum of approximately 40 seconds and the testing duration nearing 038 seconds, suggesting stable model performance and effective convergence. This performance significantly surpasses the YHTCSCA, underscoring its effectiveness in detecting zebra crossings, establishing it as the most effective method for pedestrian safety applications.

Table 4 indicates that the suggested YHTCSCA has an overall latency of 0.055s, which is slightly more than RGB-D with AI [12] (0.0115s) and AI-Sense [13] (0.0107s), but still offers an acceptable time for operation within the parameters of real-time at <1s. As compared to MobileNet

[11] (0.0136s), YHTCSCA has similar inference efficiency, and the difference in overall time is primarily due to the extra synchronization time. Overall, the proposed overall time for the YHTCSCA (0.0155 s) is longer than that of MobileNet (0.0136 s), because it added more synchronization steps to improve object detection in more complicated and variable settings. This decision means that the system includes more advanced detection networks, enhancing accuracy and robustness without compromising real-time performance. MobileNet provides more computational resources to be efficient rather than on accuracy and stability in high variance situations because the latter is not reliable enough in mobile robust applications. Moreover, the YHTCSCA offers a cost-effective option for assistive navigation and

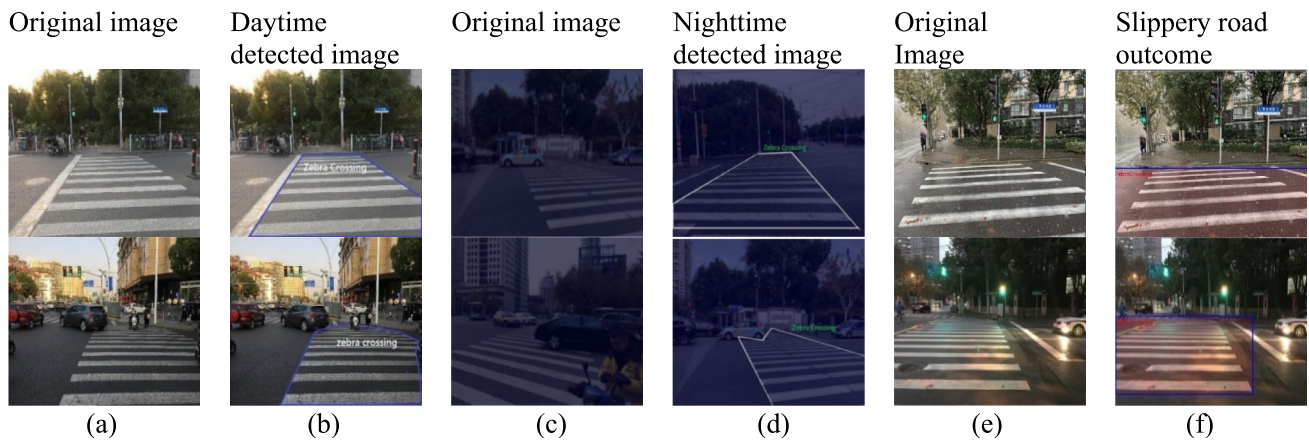
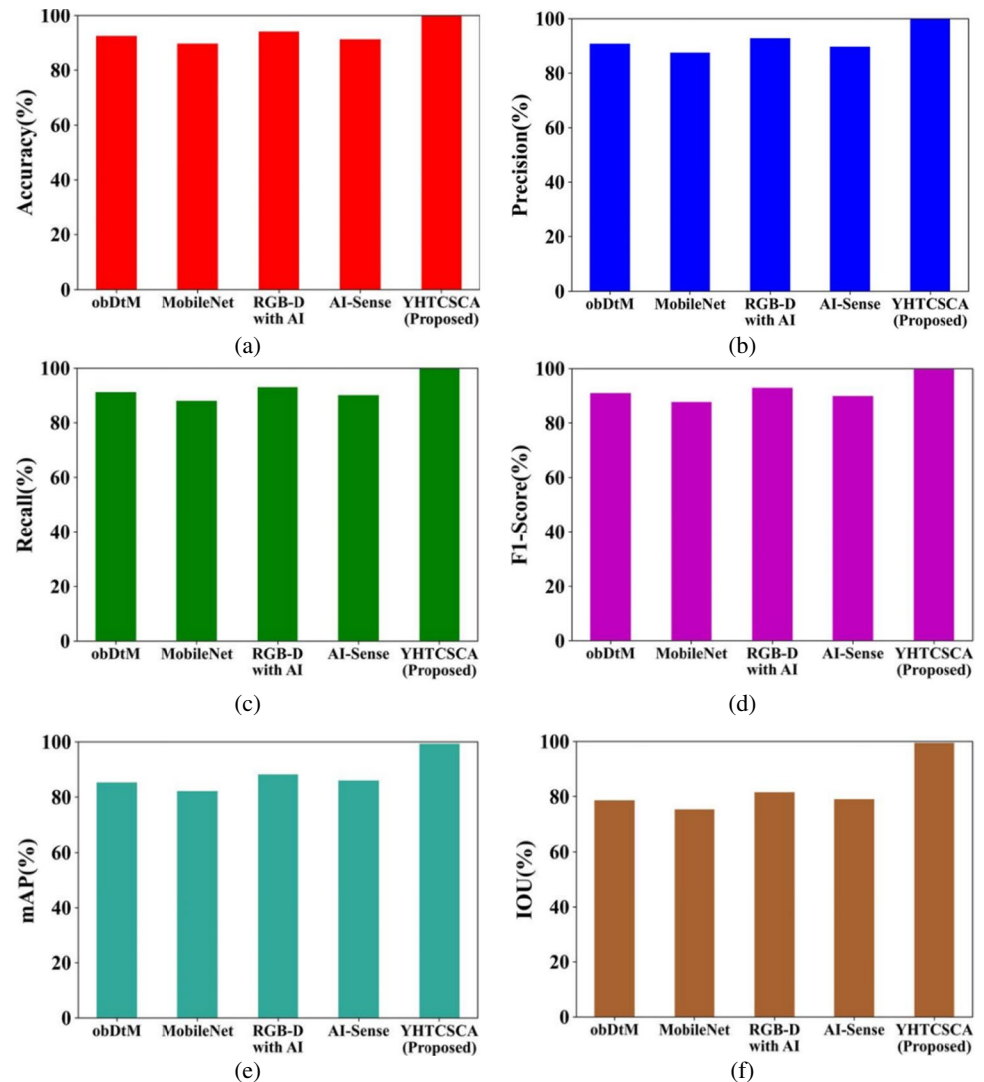


Fig. 8 Outcome based on the (a, b) Original and daytime detected image, (c, d) Original and nighttime detected image and (e, f) Original and slippery road detected image of zebra crossing

Fig. 9 Analysis of (a) Accuracy, (b) Precision, (c) Recall, (d) F1-score, (e) mAP and (f) IoU



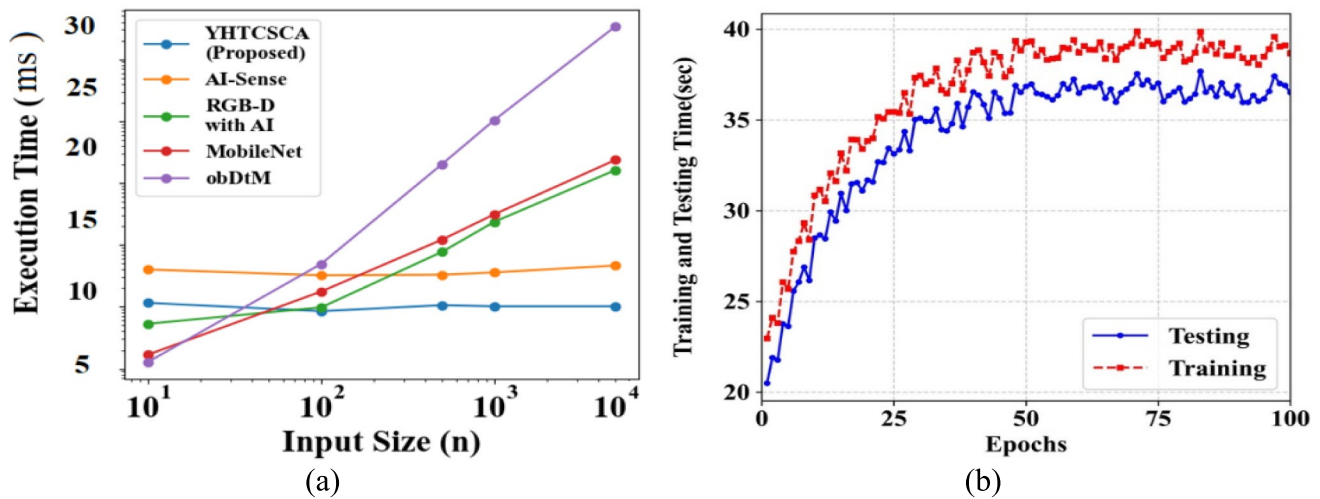


Fig. 10 Evaluation of (a) Execution time and (b) Training and testing time

Table 4 Evaluation of the time analysis

Component	Inference time (s)	Input/output and synchronization (s)	Overall time (s)
MobileNet [11]	0.0289	0.0018	0.0136
RGB-D with AI [12]	0.0064	0.0012	0.0115
AI-Sense [13]	0.0072	0.0048	0.0107
YHTCSCA (proposed)	0.0056	0.0012	0.0155

architecture environments with enhanced robust detection of real-world objects despite the slight increase in processing time for recognition.

Figure 11 presents a Receiver Operating Characteristic (ROC) curve, which demonstrates the efficacy of a binary classification approach. In contrast, the blue line represents the performance of the YHTCSCA (Proposed) model, which attains an AUC of 99.7%. This result signifies that the YHTCSCA technique demonstrates exceptional proficiency in distinguishing between the two classes, showcasing a high TPR and a low FPR, indicating superior performance.

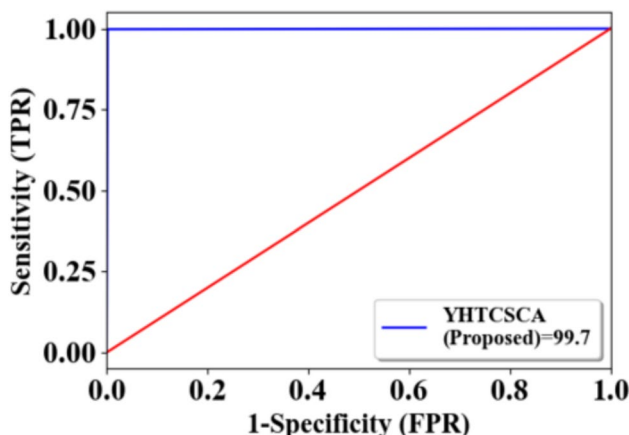


Fig. 11 Analysis of TPR and FNR

Figure 12 depicts the performance of an ObDM during training and validation across 100 epochs. The training loss, encompassing box, classification and objectless components, shows a consistent decline, signifying effective learning by the model. This model attains impressive precision and recall rates of approximately 0.98 and 0.95, respectively. Furthermore, the mAP at 0.5 IoU is recorded at 0.8, while the mAP at 0.5–0.95 IoU stands at 0.6, reflecting a strong performance in the realm of object detection.

Figure 13(a-b) illustrate the assessed functionality of the suggested object recognition system, detecting zebra crossings and alerting to the status of traffic lights. These figures illustrate four situations: (a) and (c) show zebra crossings with a stop signal for pedestrian delay, whereas (b) and (d) show zebra crossings on a green light, meaning it's a designated safe crossing point. The value of the system is based on the potential of multi-sensory information, combining visual detection with audio notifications to ensure pedestrian detours occur safely.

Figure 14(a-d) describe the evaluation curves for object detection, specifically for the class zebra crossing. The Precision-Confidence curve demonstrates a high level of precision, maintaining a value of 1.0 at a confidence level of 0.996, even at lower confidence thresholds. Lastly, the F1-Confidence curve achieves its maximum at 0.93 when the confidence level is 0.579, highlighting the optimal threshold for enhancing the F1 score.

Figure 15 depicts cumulative processing time across four pipeline stages for both optimized and actual cases. In the cases of RMSHE and HUDeMC utilizing leve-MCN, the bulk processing time is actually negligible (~0.05 min each). By contrast, while the processing time is still less than 0.5 min for detection with YHTCSCA (~0.45 min (actual) and ~0.40 min (optimized)), this increase comes from the more

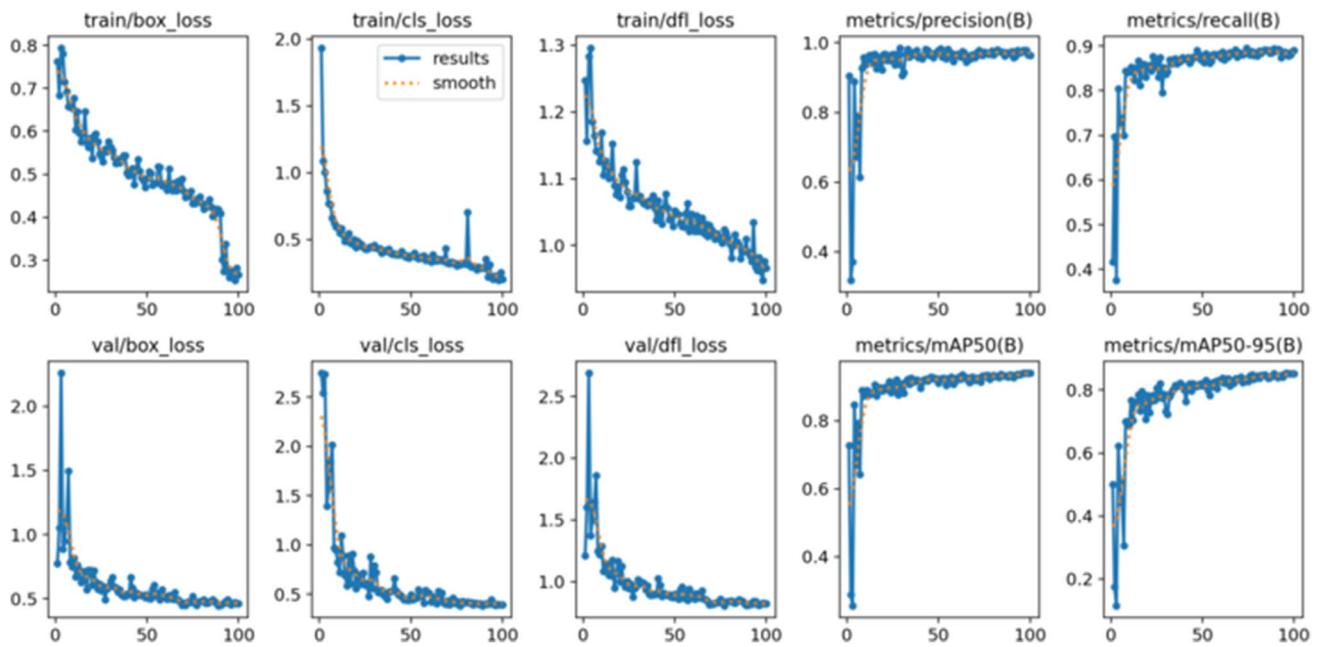
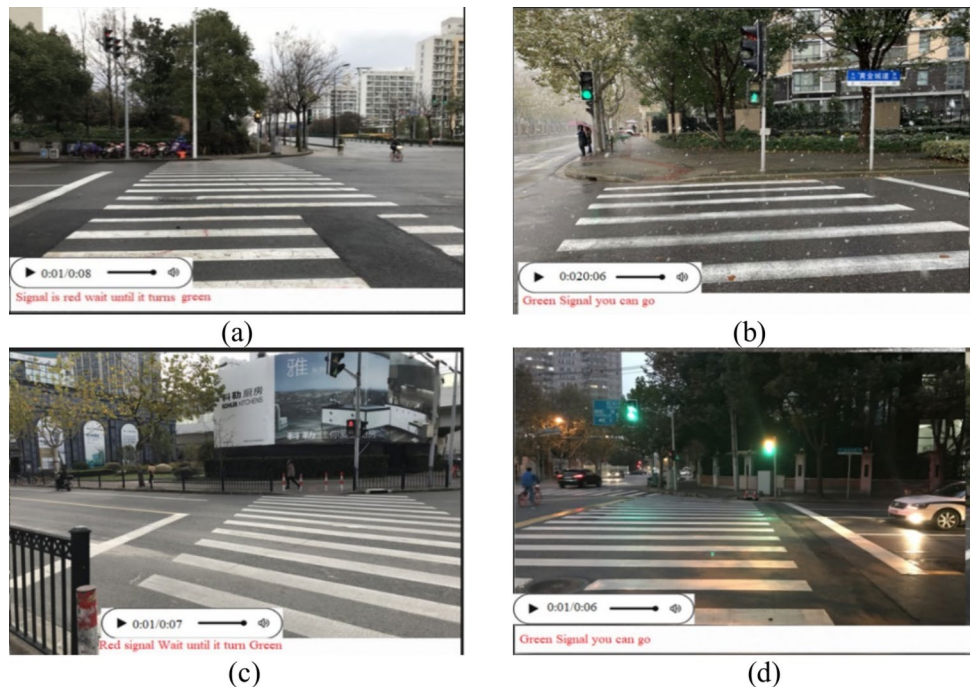


Fig. 12 Evaluation of the training and validation results

Figs. 13 (a-d) Zebra crossing detection with traffic signal (a-c) Red signal, and (b-d) Green signal



complex detection task and synchronized processing. SROA raises the total processing time again to ~ 0.5 min (actual) and ~ 0.47 min (optimized). However, the YHTCSCA and SROA are increasing processing time; the total cumulative processing time is still far less than 1.5 min. This means that relatively high accuracy and robustness to disturbances are afforded by slightly extending processing time from ~ 0.05

min to ~ 0.5 min, while still retaining the capacity for real-time processing.

Figure 16 illustrates a pair plot that depicts the interrelationships among the bounding box coordinates (x, y, width, height) utilized in object detection. The diagonal of the plot features histograms for each variable, with x and y centered approximately at 0.5, width exhibiting a peak near 0.2, and

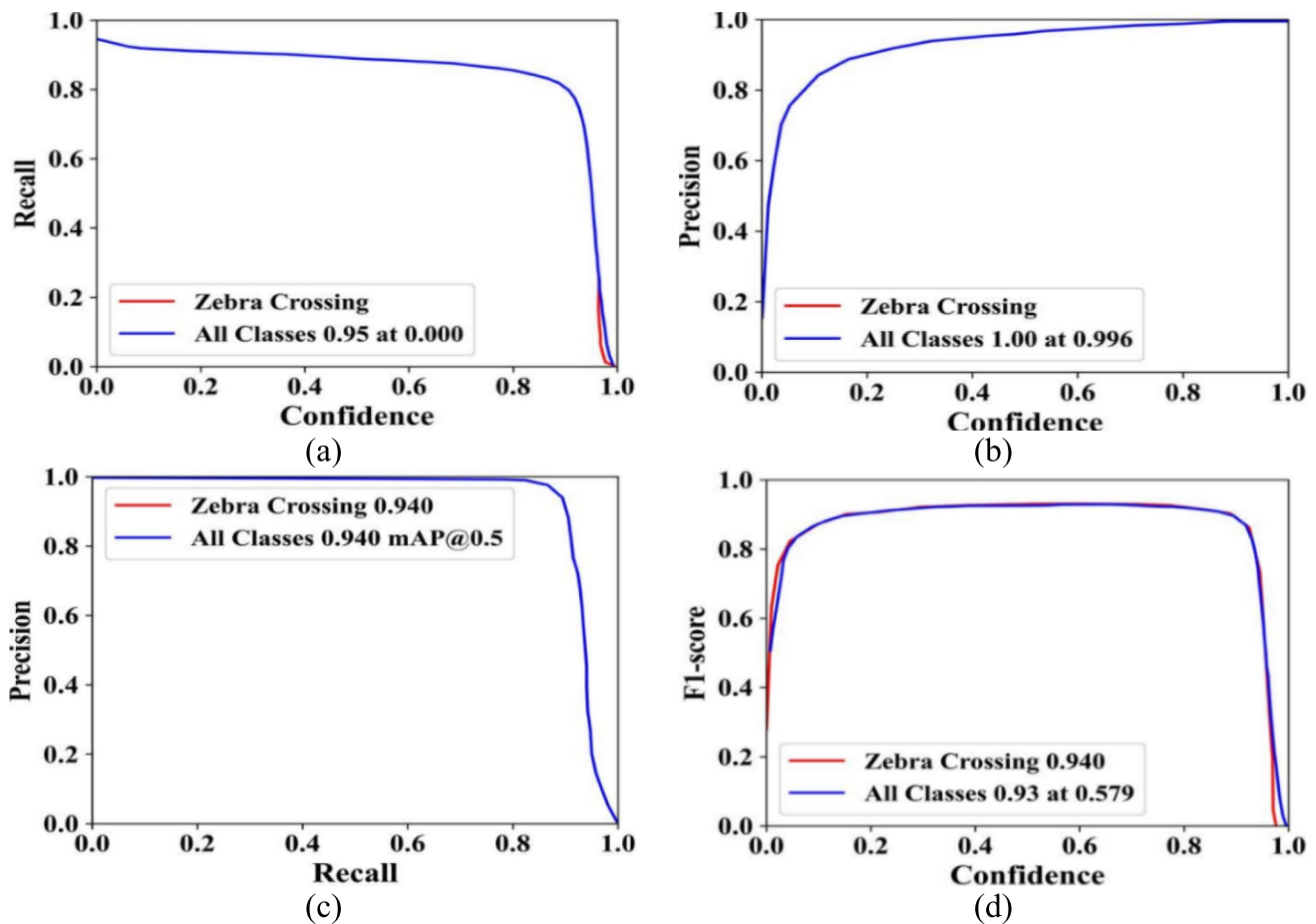


Fig. 14 Outcome based on (a) Recall confidence, (b) Precision confidence, (c) Precision recall and (d) F1 confidence curve

height peaking around 0.4. The scatterplots indicate various correlations, including a negative correlation between x and width, as well as a positive correlation between y and height.

Table 5 illustrates a 10-fold validation analysis evaluating the effectiveness of the model in terms of precision, recall, accuracy and F1-score. These outcomes establish a steady improvement across the folds, with accuracy ranging from

89.99% to 96.54%. Precision, recall, and F1-score show a similar upward trend, indicating a reliable and consistent model performance. The average metrics are as follows: accuracy (99.79 %), precision (99.82 %), recall (99.81%), and F1-score (99.76%), highlighting the model's strong predictive capability and balanced performance across different measures.

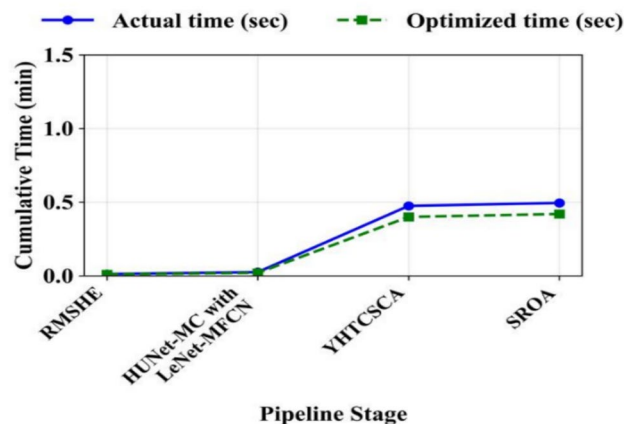


Fig. 15 Analysis of cumulative time

Objective Solution

This section studies the effectiveness of the SROA approach by contrasting it with a number of standard approaches and shows how this step improves training stability through more accurate parameterization of the loss function.

Figure 17 demonstrates that the algorithms Improved Chimp Optimization Algorithm (ICOA) [34], Parrot Optimizer (PO) [35], Greylag Goose Optimization (GGO) [36], and the suggested approach exhibit significant convergence rates. In contrast, the newly developed NSO algorithm demonstrates a lower convergence rate to local optima, which is approximately 1.5% greater than that of the EWOA algorithm.

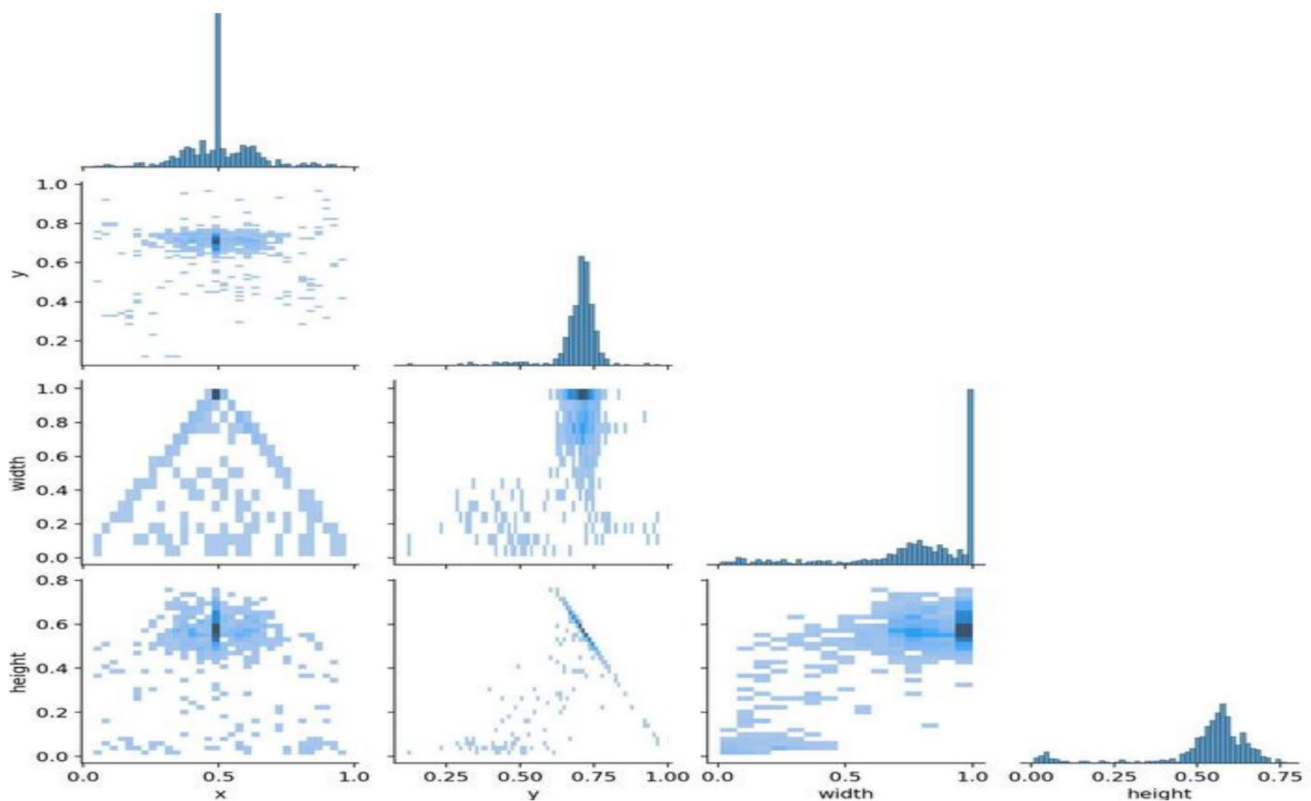


Fig. 16 Analysis of the interrelationship between pair plot

Table 5 K-fold validation analysis

Folds	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Fold 1	89.99	87.45	89.97	87.90
Fold 2	89.95	89.34	90.89	89.45
Fold 3	90.97	90.45	90.95	91.24
Fold 4	90.96	91.90	91.99	92.56
Fold 5	91.99	91.56	91.96	91.89
Fold 6	91.95	92.89	92.97	93.9
Fold 7	92.98	93.90	93.98	92.34
Fold 8	94.97	94.78	94.97	94.56
Fold 9	95.99	95.89	96.98	95.89
Fold 10	96.54	96.98	97.39	96.67
Mean value	99.79	99.82	99.81	99.76

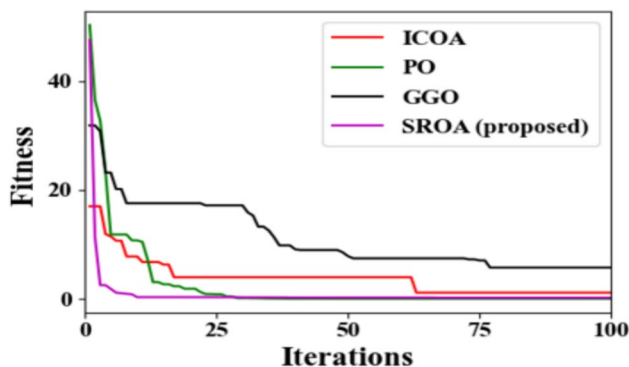


Fig. 17 Evaluation of the best fitness function

Table 6 presents an evaluation of various optimization algorithms, detailing their performance metrics compared with ICOA [34], PO [35], GGO [36] and SROA (proposed). The proposed SROA algorithm outperformed the others, achieving an optimum value of 0.97, a worst value of 0.87, a mean of 0.92, a standard deviation of 0.05, and the shortest computation time of 10.2 seconds.

Analysis of the Ablation Experiment

This analysis provides a comparative valuation of various formations of the YHTCSA technique by investigating their performance against key metrics.

Table 7 evaluates the impact of removing fundamental components of the proposed YHTCSA model. The proposed YHTCSA yields the best performance, achieving

Table 6 Assessment of current algorithms with suggested approach

Methods	Optimum value	Worst value	Mean	Standard deviation	Computation time (seconds)
ICOA [34]	0.92	0.71	0.84	0.06	17.2
PO [35]	0.89	0.68	0.78	0.08	18
GGO [36]	0.94	0.73	0.86	0.10	15.3
SROA (proposed)	0.97	0.87	0.92	0.05	10.2

Table 7 Evaluation of ablation experiment

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Without Haze U-Net	95.4	93.65	97.5	94.7
Without LeNet-MFCN	97.6	97.1	96.4	96.5
Without YOLOv11	95.2	87.1	87.1	87.5
Without CNN-ECAM	89.6	78.7	75.4	76.1
Proposed YHTCSCA	99.7	99.8	99.8	99.7

Table 8 Analysis of ablation experiment on YHTCSCA (proposed)

Learning rate	Detected accuracy (%)
0.001	91.5
0.002	92.9
0.003	96.3
0.004	99.3
Batch size (samples)	Detected accuracy (%)
4	90.8
8	93.9
16	97.4
32	98.1
64	99.4

99.7% for accuracy and 99.8% for precision, recall, and F1 score. Upon removing LeNet-MFCN, there is some performance loss, resulting in an accuracy of 97.6%. In contrast, the exclusion of Haze U-Net or YOLOv11 results in more substantial reductions, with accuracies of 95.4% and 95.2%, respectively. The most pronounced decline is observed when CNN-ECAM is removed, leading to an accuracy of 89.6%, a recall of 75.4%, and an F1-score of 76.1%, underscoring its vital importance to the model's overall effectiveness.

Table 8 summarizes the models used in the ablation analysis of the presented YHTCSCA to show the impact of learning rate and batch size on the accuracy of prediction. This increases the learning rate from 0.001 to 0.004, improving accuracy from 91.5% to 99.3%, demonstrating that utilizing a learning rate of 0.004 allows for better convergence. Likewise, raising the batch size from 4 to 64 boosts accuracy from 90.8% to 99.4%, implying that larger batch sizes contribute to enhanced model stability and performance. These findings underscore the importance of adjusting the hyperparameter to achieve optimal detection accuracy.

Final Remarks

The YHTCSCA methodology proficiently identifies traffic lights, objects, and pedestrian crossings through the use of Python, thereby improving safety for pedestrians, especially those with visual impairments. This approach utilizes edge detection and classification methods to ensure precise signal

identification and navigation assistance. This compares current approaches like ObDtM, MobileNet, and AI-SenseVision approaches with YHTCSCA, showing the proposed model has greater accuracy, precision, recall, and time efficiency. The approach with an execution time of 0.0001 s at an AUC of 99.7% is believed to have the greatest classification power. The training showed a consistent reduction of loss and an increase in mAP, meaning the proposed model is reliable for object detection. A convergence analysis of optimization algorithms gives evidence for the efficiency of NSO in avoiding local optimums.

Limitations

In the YHTCSCA approach, while demonstrating high accuracy and employing sophisticated techniques, performance is compromised in highly dynamic settings characterized by a level of computational complexity that hinders processing capabilities on low-power edge devices. Moreover, challenging weather conditions, such as heavy rain or fog, limit a system's efficacy for haze removal and object detection. This reliance of the system on pre-trained models injects biases that influence the ability of the system to generalize over various urban settings. Additional improvements are necessary to enhance its robustness and adaptability in a range of real-world situations.

Case Investigation

A real-world case experiment is conducted in a busy city with visually impaired participants using the proposed system, which integrates MZCDFVIN for traffic detection, YHTCSCA with YOLOv11 for obstacle detection, and an updated CNN for zebra crossing line detection (using Hough Line Transform) across various environments. Environmental challenges like haze, rain, and low-light conditions are addressed with HUNet-MC, which effectively reduces haze, while local contrast is enhanced using RMSHE. SROA is included to ensure the model's stability and account for latency caused by HUNet-MC processing. The case experiments are collected in real-time responses through vibration and short audio cues from the device. Results showed a 98.2% accuracy in detecting crossings, 96.8% in obstacle detection, and a 97% success rate for users navigating successfully. This research demonstrated that the proposed solution operates in real time, is robust, and surpasses previous untested audio-based systems in real-world environments.

This approach is better because, unlike the previous method, which mainly depended on untested audio signals and lacked real-world testing, the suggested approach has full situational awareness through YOLOv11 detection of pedestrians, vehicles, and obstacles in real time.

Additionally, this approach improves visibility in haze, shadows and low-light conditions with HUNet-MC and RMSHE. Moreover, this approach implemented a robust optimization routine using the SROA method to prevent training instabilities and performance drops. The practical approach combines advanced DL and image enhancement techniques to operate effectively in real-world environments instead of limited, controlled settings, resulting in a more reliable and safer experience for blind users.

Conclusion

This research employed a newly developed framework called YHTCSCA for the classification of traffic lights and the detection of a zebra crossing. For the visually impaired, individuals with impaired vision, this approach identifies obstacles and locates zebra crossings in places that allow for safe navigation. This meets the different levels of difficulty with a zebra crossing-based solution. The assessment of performance is done through an analysis of optimization results alongside performance metrics and results, which improved the classification and detection of zebra crossings. The suggested method shows excellent performance on all metrics evaluated with an accuracy of 99.79%, precision of 99.82%, recall of 99.81%, F1-score of 99.76%, mAP of 99.3% and IoU of 99.5%. Future scope focuses on advanced AI-driven assistance technologies for individuals with visual impairments that incorporate Augmented Reality (AR) navigation through XR glasses, AI-enhanced smart canes equipped with LiDAR and ultrasonic sensors, and haptic feedback devices that provide immediate alerts for obstacles. Speech-based AI companions, utilizing Explainable AI (XAI), deliver intuitive navigational support. Furthermore, TinyML-enabled Edge AI facilitates low-latency processing for object recognition and hazard identification. This also encapsulates Multi-Modal Sensing (MMS), Edge-AI Optimization (EAIO), adaptive learning approaches, Internet of Things (IoT) smart city connectivity, and large-scale trials to strengthen the robustness, personalization, and real-world usability of a navigation system for visually impaired end users. Moreover, 5 G-enabled cloud AI offers traffic analysis and predicts pedestrian movements, thereby enhancing safety at crossings. These advancements greatly enhance the autonomy, security, and confidence of VIB individuals in urban settings.

Authors' Contribution Both the authors have contributed equally to the work.

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Data Availability The data supporting the findings of this study are cited and the link is listed in the references.

Declarations

Ethical Approval All applicable institutional and/or national guidelines for the care and use of animals were followed.

Informed Consent For this type of analysis formal consent is not needed.

Conflict of interest The authors declare no competing interests.

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